

ESS_2

gitivan

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Preface

This is a Quarto book.

1 Lecture 1: Hank model

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3 Summary

In summary, this book has no content whatsoever.

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References

9 f_X(x)

$$\frac{\theta 2^\theta}{x^{\theta+1}} \mathbf{1}_{[2,+\infty)}(x)$$

9.1 (a) Find a complete sufficient statistic

Solution:

$S = \sum_i \ln X_i$ is complete and sufficient by the property of a full-rank exponential family.

9.2 (b) Find MLE $\hat{\theta}$

Solution:

$$(1) \quad \ell(\theta) = n \ln \theta + n\theta \ln 2 - (\theta + 1) \sum_i \ln x_i$$

$$(2) \quad \frac{\partial \ell(\theta)}{\partial \theta} = 0 \quad \Rightarrow \quad \hat{\theta}_{\text{MLE}} = \frac{n}{\sum_i \ln(X_i/2)}$$

9.3 (c) Is $\hat{\theta}$ unbiased? + MSE

Solution:

(1) Find the distribution of $Y = \ln X$

(1.1) Intermediate calculations:

$$\begin{aligned}
 x &= e^y, & \frac{\partial x}{\partial y} &= e^y \\
 f_Y(y) &= f_X(e^y) \left| \frac{\partial x}{\partial y} \right| \\
 &= \frac{\theta 2^\theta}{e^{y(\theta+1)}} e^y \mathbf{1}_{[\ln 2, +\infty)}(y) \\
 &= \theta 2^\theta e^{-\theta y} \mathbf{1}_{[\ln 2, +\infty)}(y) \\
 &\Rightarrow Y \sim \text{Ga}(1, \theta)
 \end{aligned}$$

(2) Distribution of $\sum_i Y_i = S$:

$$S \sim \text{Ga}(n, \theta)$$

(3) Expectation of S^{-1} :

$$\begin{aligned}
 \int_{2n \ln 2}^{\infty} s^{n-2} e^{-\theta s} ds &= C \times \underbrace{\int_{2n \ln 2}^{\infty} A s^{n-2} e^{-\theta s} ds}_{=1, \text{ since this is a density}} \\
 \text{where } C &= \frac{\Gamma(n-1)}{\Gamma(n)} \frac{\theta^n}{\theta^{n-1}} = \frac{\theta}{n-1}
 \end{aligned}$$

(4) Final expression:

$$\mathbb{E}[\hat{\theta}_{\text{MLE}}] = \frac{n\theta}{n-1}$$